Where the Dodecahedron Goes

The Cleo star, the DICTO Metaxyhexecontahedron and the DICTO Palintriacontahedron: a companion to Progenitors



DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com
Working draft · 17 October 2026 · companion to Progenitors (DICTO DOI: 10.5281/zenodo.23413184) · DICTO DOI: on publication · Licence CC BY 4.0 · Data, scripts, exact checks and renders: https://github.com/DICTOR-Master/kaleidohedra (branch research: studies/2026-10-16-progenitors)
The Cleo star with the dodecahedron in it, the inward pyramids, the full-height icosidodecahedron star, the solid in the last gap and all names are DICTO's; the exact computations, searches and text were done with an AI assistant (Claude, Anthropic); a second model (Fable) advised on scope and publishing order.

Abstract. Progenitors (DICTO DOI: 10.5281/zenodo.23413184) set the DICTO Ogdoecontahedron O into the Spheres of Dharma in the dodecahedron's place and found an exact nine-station golden cycle, I ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ I·φ³, closing on φ⁹ in volume. This companion asks where the dodecahedron went and what fills the cycle's loose steps. Every solid here is exact in ℚ(√5), and every nesting and volume is checked with remainder 0 unless marked float. The dodecahedron returns inside the icosidodecahedron, as the hull of DICTO's Cleo star: the icosahedron's 20 Cleo-Icos with 20 more turned outward have their tips exactly at (2/3)·D, and 82 pieces of four kinds fill that dodecahedron exactly. The loose step from the rhombic triacontahedron to the next icosahedron is closed by the **DICTO Metaxyhexecontahedron**: the member of the dodecahedron–icosahedron hull family whose icosahedron edge is 9/φ⁴ times its dodecahedron edge. It touches both neighbours, both of its layers are a single kind of piece, and its dual closes the same step in the polar-reversed cycle. The full-height icosidodecahedron star gives a second member of the same family, the **DICTO Palintriacontahedron**, the rhombic triacontahedron turned inside out. The cycle grows to twelve stations; the loose step O ⊂ rTI stays open. Prior-art status is stated claim by claim in §5.

1. Where Progenitors left off

The DICTO Truncated-Icosahedral Hecatodihedron H (DICTO DOI: 10.5281/zenodo.23360857), 102 faces, and its polar, the DICTO Ogdoecontahedron O (DICTO DOI: 10.5281/zenodo.23404688), 80 faces, both came out of the Spheres of Dharma (DICTO DOI: 10.5281/zenodo.23310833). Progenitors (DICTO DOI: 10.5281/zenodo.23413184) showed that on their own the pair never close golden, but that with O on the Dharma's own icosidodecahedron at scale 3φ²/4, in the dodecahedron's place, every step of the following cycle nests exactly and the cycle closes on I·φ³ / I = 38 + 17√5 = φ⁹ in volume:

I ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ I·φ³

We write I, ID, D, TI, PD, RT for the icosahedron, icosidodecahedron, dodecahedron, truncated icosahedron, pentakis dodecahedron and rhombic triacontahedron, all in the Dharma's sizes (icosahedron edge 1/φ); rTI is the rectified truncated icosahedron and jtI its dual. Table 1 repeats the contacts. Three steps are loose: I ⊂ ID, O ⊂ rTI and RT ⊂ I·φ³. The dodecahedron itself is gone, since O and D cross. This paper brings the dodecahedron back in

the first loose step (§2) and closes the last one with a new solid (§3); the middle one stays open.

Step	Contact (exact)
I ⊂ ID	loose (0 of 12 corners on ID's faces)
ID ⊂ O	tight: all 30 corners of ID on faces of O
O ⊂ rTI	loose
rTI ⊂ TI	tight: all 90 corners of rTI on TI's surface (its edges)
TI ⊂ H	tight: all 60 corners of TI on faces of H
H ⊂ jtI	tight: all 80 corners of H on faces of jtI
jtI ⊂ RT	tight: 80 corners of jtI on faces of RT
RT ⊂ I·φ³	loose (0 of 32 corners on faces of I·φ³)

Table 1. The nine-station cycle of Progenitors (exact), repeated from its Table 1.

2. The dodecahedron returns: the Cleo star (exact)

With O in D's place, DICTO felt that "the dodecahedron should be in there too". The Dharma icosahedron is 20 Cleo-Icos, pyramids from the centre over its faces (Theaetetus, DICTO DOI: 10.5281/zenodo.23378070). Turning 20 more outward as spikes makes the Cleo star (volume 2·I). **Its 20 tips are exactly the corners of (2/3)·D**, the Dharma dodecahedron at two thirds. That the tips form a dodecahedron is elementary, since they lie on the three-fold axes at one radius; the size is the Dharma-specific fact. This dodecahedron D′ fits loosely inside ID and inside O, so the dodecahedron returns in the old loose step I → ID:

$$\mathbf{I \subset \text{Cleo star} \subset D' \subset ID \subset O \subset rTI \subset TI \subset H \subset jtI \subset RT \subset I \cdot \varphi^3}$$

D′ filled exactly. The gap D′ − star is one connected region (float flood fill). DICTO placed 12 inward pentagonal pyramids in it, one on each face of D′ with its apex at the icosahedron corner below. What is left is 30 valley tetrahedra, one along each edge of I, with corners at the edge's two ends and at the tips of the two spikes beside it. All 30 are one shape. The 82 pieces of four kinds fill D′ with remainder 0 (Table 2, Figure 1). The Cleo star is not a stellation of the icosahedron: its spike faces lie in no face plane of I. It is the triakis icosahedron (a Kleetope of I) at one particular height.

Piece	Count	Volume in all (exact)
Cleo-Ico, core (the icosahedron)	20	$-5/12 + 5\sqrt{5}/12$
Cleo-Ico, spike	20	$-5/12 + 5\sqrt{5}/12$
inward pentagonal pyramid	12	$5/9 - \sqrt{5}/27$
valley tetrahedron	30	$25/18 - 5\sqrt{5}/18$
D′ = (2/3)·D	82	**$10/9 + 14\sqrt{5}/27$** (remainder 0)

Table 2. The Cleo star build of D′.

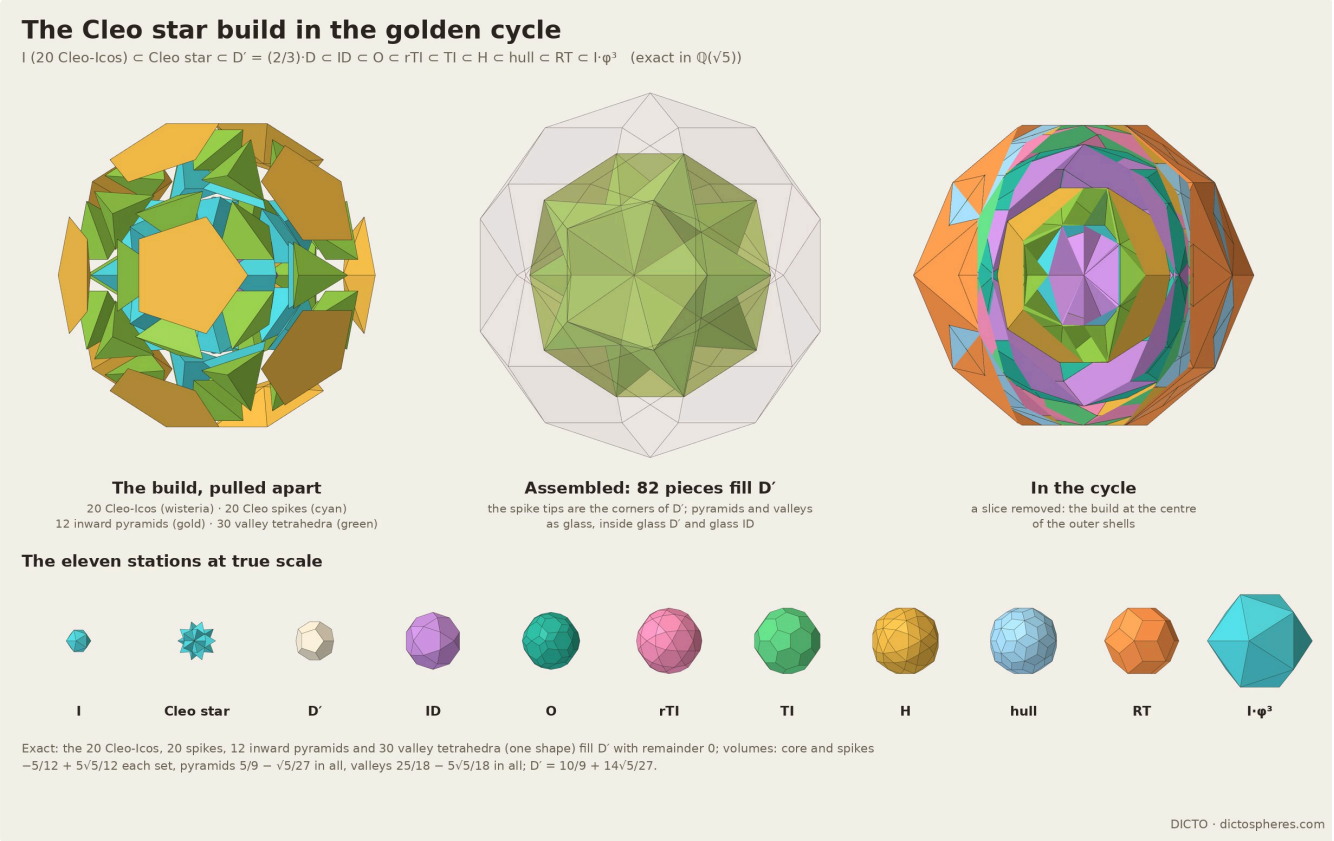


Figure 1. The Cleo star build: pulled apart, assembled inside glass D′ and ID, and at the centre of the cycle; the eleven stations at true scale. Drawn from exact pieces.

3. The last loose step: the DICTO Metaxyhexecontahedron (exact)

In the Spheres of Dharma the triacontahedron reaches only 0.812 of the way to the next icosahedron's faces. Looking at the cycle, DICTO judged that one more solid should fit there. Among the convex icosahedral solids, none touches both RT and I·φ³ (Table 3, float). The dodecahedron–icosahedron hull family, however, contains exactly one member that does. It is the convex hull of the RT's 12 five-fold corners and the 20 face centres of I·φ³. DICTO named it the **DICTO Metaxyhexecontahedron** M, from the Greek *metaxý*, "in-between", and 60 faces.

- **Shape.** 32 corners (12 five-fold, 20 six-fold), 90 edges and 60 congruent isosceles triangles: the same combinatorics as the pentakis dodecahedron. Its icosahedron edge is exactly **9/φ⁴** times its dodecahedron edge (squared, 81(47 − 21√5)/2 = 81/φ⁸), ≈ 1.31308. The family is fully symmetric for ratios from (5 + 3√5)/10 to φ. Its named members are the Catalan pentakis dodecahedron, at 3(7 + 5√5)/38 ≈ 1.435, and the DICTO Palintriacontahedron of §4, at 3/√5 ≈ 1.342.
- **Fit.** The RT lies inside M, and its 12 five-fold corners are M's own. M lies inside I·φ³, and its 20 three-fold corners sit on the centres of I·φ³'s faces. In Dharma units the squared corner radii are 45/8 − 9√5/8 and 47/24 + 7√5/8. M has an insphere (all 60 faces at one distance) but no midsphere, because its two kinds of edge sit at different distances.
- **Pieces.** Cut as in *Inverted Shells* (one piece per face of the inner solid, cut by the cone over that face), both new layers are a single kind of piece. RT → M gives 30 pieces, each with 6 corners (a rhombus and 6 triangles) and volume 353/72 − 149√5/72. M → I·φ³ gives 60 pieces, each with 5 corners and 6 triangles (a triangular bipyramid shape) and volume 1/8 + √5/18. vol M = 145/12 + 65√5/12, and M takes 0.356 of the step's volume.

- **Its dual.** Because M has an insphere, its polar has all 60 corners on one sphere. It has 12 regular pentagons and 20 hexagons whose sides alternate in the ratio $(14 + 9\sqrt{5})/11 \approx 3.102$: an icosahedron cut by a dodecahedron, a truncated icosahedron of non-Archimedean depth. In the ends sphere of *Double Vision* ($\rho^2 = \varphi^2/4$), it sits between the polars of $I{\cdot}\varphi^3$ and RT. All 60 of its corners lie on the faces of polar(RT), and the 20 corners of polar($I{\cdot}\varphi^3$) touch its hexagons. So it closes the matching loose step of the reversed cycle (exact).
- **An observation.** $9/\varphi^4$ is also the exact gain per round of the tightest progenitor 3-cycle in Progenitors (§2 there). We note the coincidence without an explanation.

Solid, just big enough to hold RT	Reach toward $I{\cdot}\varphi^3$'s faces (float)
Hecatodihedron	0.9948
icosidodecahedron; Palintriacontahedron	0.9787
deltoidal hexecontahedron	0.9443
truncated dodecahedron	0.9325
pentakis dodecahedron	0.9149
rhombicosidodecahedron	0.8754
truncated icosahedron	0.8682
dodecahedron; triakis icosahedron	too big
DICTO Metaxyhexecontahedron	**1 (exact)**

Table 3. Candidates for the RT → $I{\cdot}\varphi^3$ step. 1 means touching both neighbours.

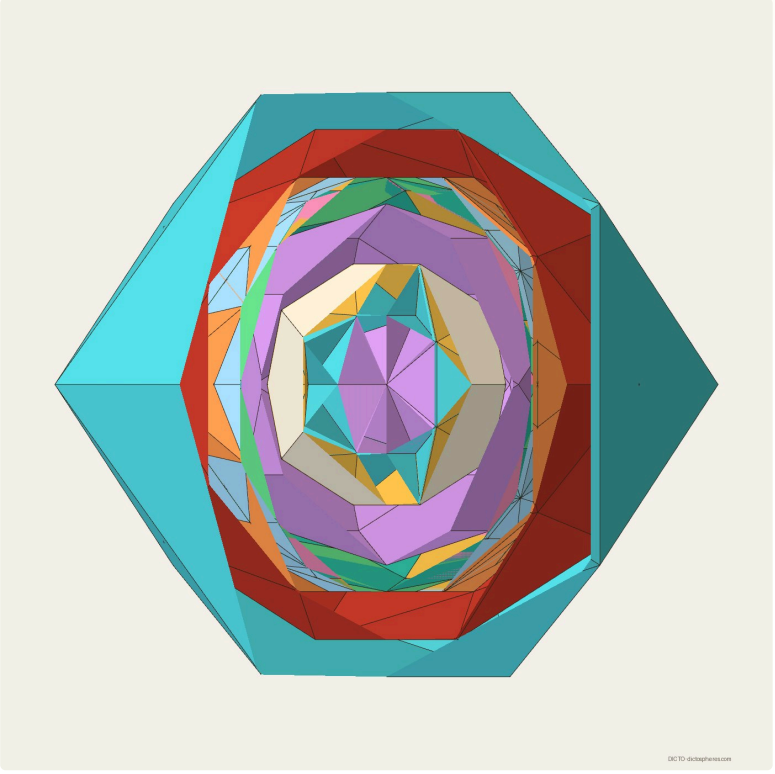


Figure 2. The DICTO Metaxyhexecontahedron (brick red) in the cycle, between the rhombic triacontahedron (orange) and the next icosahedron (cyan); a slice facing the viewer removed. Drawn from exact stations.

4. The seam: the DICTO Palintriacontahedron

DICTO also flipped the icosidodecahedron's own central pyramids out at full height ("shortening feels like cheating; one more solid is needed"). The hull of the tips is the **DICTO**

Palintriacontahedron X (Greek *pálin*, back): 32 corners and 60 faces, each half a rhombic-dodecahedron rhombus ($\cos^2$ of the face angles 1/9, 1/3, 1/3). Its icosahedron edge is exactly $3/\sqrt{5}$ times its dodecahedron edge (squared, 9/5). X is the rhombic triacontahedron turned inside out: it has the same 32 axes, with the reciprocal radius ratio. It belongs to the same family as M. As a float observation only: X lies inside the next turn's D′, and 99.46% of its volume (20,000 sample points) lies inside the next icosahedron's Cleo star, so it bridges one turn to the next. As a further float observation, the first stellation of the icosahedron puts its tips at $3/\sqrt{5}$ times the face distance, the same number. Figure 3.

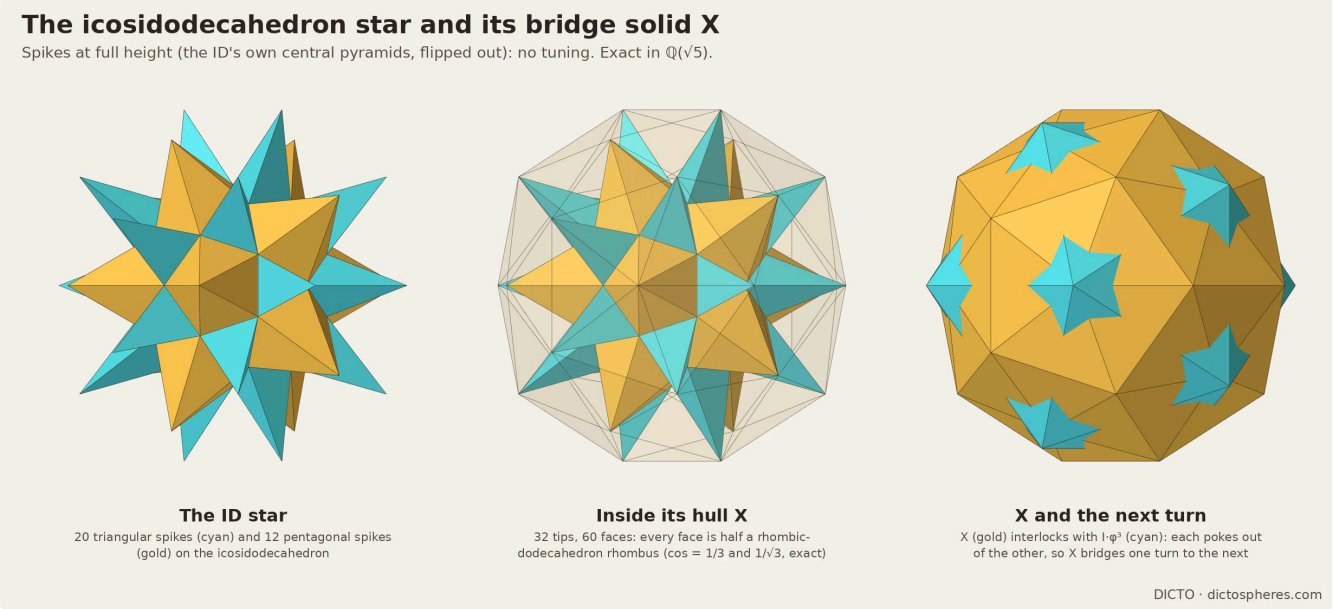


Figure 3. The icosidodecahedron star and its hull X, the DICTO Palintriacontahedron; X interlocked with the next icosahedron. Drawn from exact corners.

5. Relation to known work, and status

- **Known, cited, not claimed.** Nested Platonic solids go back to Kepler (1596), and nested icosahedral shells appear in virology and fullerene work (Dechant et al. 2014). The pentakis dodecahedron as the convex hull of a dodecahedron and an icosahedron, and its fully symmetric range, are on the Polytope Wiki. The Catalan member and the biscribed member are listed by MathWorld and McCooey. The truncated icosahedron–pentakis dodecahedron compound, the rectified truncated icosahedron and its dual jtI are classical (MathWorld; Wikipedia; McCooey). The Kleetope construction is from Grünbaum. The Cleo-Icos are Hypsicles' central pyramids (Euclid, *Elements*, "Book XIV"). Polar reciprocation is classical.
- **Not found, in the sources checked:** the exact four-kind filling of (2/3)·D by the Cleo star with inward pyramids and valley tetrahedra was not found either. The Metaxyhexecontahedron and the Palintriacontahedron were not found as named members of the hull family, nor the $9/\varphi^4$ and $3/\sqrt{5}$ members as distinguished. The names "Metaxyhexecontahedron" and "Palintriacontahedron" return no results. (The golden cycle itself and the H–O compound are claimed, with their search, in Progenitors.)
- **Searched (16 October 2026).** Wikipedia, MathWorld, the Polytope Wiki, Klitzing, OEIS, McCooey's Visual Polyhedra, G. Hart's *Virtual Polyhedra* (index, compound list and the full alphabetic model list: compounds of the uniform solids with their duals, the rhombic enneacontahedron, the pentakis dodecahedron; none of the solids or cycles here), Wenninger's model lists, Wassell and Benito (2012) and Bridges 2017 (dual-pair

edge ratios; the former found the dodecahedron–icosahedron ratio φ : 3 absent from the literature until 2012), arXiv (nested golden shells, polar sequences, dodecahedral dissections), and the DICTO reference library.
- **Antiprism** (A. Rossiter, version 0.29), run 16 October 2026: it recomputes the solids independently, and a sweep of every Conway form with up to two operators on I and D (840 forms) matches neither H nor O, while M, X and M's dual have the structures stated here.
- **Not read.** Wenninger's *Polyhedron Models* and *Dual Models* in print; Coxeter, Cromwell and Holden in print. The Stella catalogue was not run. Every "not found" above is limited to the sources listed; the full log is in the study (SEARCH-LOG.md).

6. Conclusion

Progenitors put the DICTO Ogdoecontahedron in the dodecahedron's place and closed the Spheres of Dharma golden again. Nothing is lost by that. The dodecahedron comes back as the hull of the Cleo star, exactly at two thirds, inside the first loose step, and is filled by 82 pieces of four kinds. The step the Dharma left open, from the triacontahedron to the next icosahedron, is closed by a single solid that touches both sides, the DICTO Metaxyhexecontahedron; both of its layers are a single kind of piece, and its dual does the same for the reversed cycle. The cycle grows to twelve stations, I ⊂ Cleo star ⊂ D′ ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ M ⊂ I·φ^3, and repeats; the full-height icosidodecahedron star adds the DICTO Palintriacontahedron, a second member of M's family, as a bridge between turns (float). **Open:** whether the loose step O ⊂ rTI has a similar closing solid; the $9/\varphi^4$ coincidence; and a physical build of the Metaxyhexecontahedron's two layers.

7. Reproducibility

In studies/2026-10-16-progenitors: `o_dharma.py`, `core_hull_in_cycle.py` (the cycle of §1, from Progenitors); `cleo_star_station.py`, `inward_pyramids.py`, `progenitors_exact.py` (§2, exact → `progenitors-exact.txt`); `gap_member.py`, `gap_chain.py`, `gap_pieces_exact.py`, `metaxy_dual.py` (§3); `id_star.py` (§4). Exact arithmetic is in $\mathbb{Q}(\sqrt{5})$ (numbers a + b√5, a and b rational), with exact sign tests for every inclusion. The figures come from `render_cleo.html` (composed by `cleo_figure.py`), `render_full.html` and `render_idstar.html`. The full record, with every step and its corrections, is in the study README and SEARCH-LOG.md.

References

1. DICTO, Spheres of Dharma: a second endless golden cycle of the icosahedral solids, through the icosidodecahedron, the fullerene cage and the rhombic triacontahedron, DICTO DOI: 10.5281/zenodo.23310833.
2. DICTO, The DICTO Truncated-Icosahedral Hecatodihedron (Version 2), DICTO DOI: 10.5281/zenodo.23360857.
3. DICTO, The DICTO Ogdoecontahedron, DICTO DOI: 10.5281/zenodo.23404688.
4. DICTO, Progenitors: the DICTO Hecatodihedron and the DICTO Ogdoecontahedron as a twin pair, DICTO DOI: 10.5281/zenodo.23413184.
5. DICTO, Inverted Shells: The Spheres of Dharma Turned Inside Out, DICTO DOI: 10.5281/zenodo.23405029.
6. DICTO, Double Vision – Both Cycles in One Space, DICTO DOI: 10.5281/zenodo.23407106.

7. DICTO, Theaetetus: an icosahedron held exactly by its Cleo-Icos, grown to the DICTO Hecatodihedron in two exact stories, DICTO DOI: 10.5281/zenodo.23378070.
8. DICTO, AXE, CLEO and FUJI: the Gap Pieces of the Icosahedron, DICTO DOI: 10.5281/zenodo.23351291.
9. DICTO, Capturing the In-betweens: a Study of Transitional Polyhedral Shells, DICTO DOI: 10.5281/zenodo.23345502.
10. J. Kepler, *Mysterium cosmographicum*, Tübingen, 1596.
11. Euclid, *Elements*, Book XIII (tr. T. L. Heath, 1908); Hypsicles, "Book XIV".
12. B. Grünbaum, *Convex Polytopes*, Wiley (1967) (Kleetopes).
13. J. H. Conway, H. Burgiel, C. Goodman-Strauss, *The Symmetries of Things*, A K Peters (2008) (Conway polyhedron notation).
14. Polytope Wiki, Pentakis dodecahedron, https://polytope.miraheze.org/wiki/Pentakis_dodecahedron.
15. E. W. Weisstein, Pentakis Dodecahedron; Truncated Icosahedron–Pentakis Dodecahedron Compound, MathWorld, https://mathworld.wolfram.com.
16. D. McCooey, Visual Polyhedra: Archimedean–Catalan Hulls, Rectified Archimedean Solids, Biscribed solids, https://dmccooey.com/polyhedra (accessed 16 October 2026).
17. Rhombic enneacontahedron (Conway jtI, dual of the rectified truncated icosahedron), Wikipedia, https://en.wikipedia.org/wiki/Rhombic_enneacontahedron; Rectified truncated icosahedron, https://en.wikipedia.org/wiki/Rectified_truncated_icosahedron.
18. G. W. Hart, *Virtual Polyhedra*, https://www.georgehart.com/virtual-polyhedra/vp.html (alphabetic list read 16 October 2026).
19. A. Rossiter, Antiprism, polyhedron modelling software, version 0.29, https://www.antiprism.com.
20. P.-P. Dechant, J. Wardman, T. Keef, R. Twarock, Viruses and fullerenes: symmetry as a common thread?, *Acta Cryst. A* 70 (2014) 162–167, doi:10.1107/S2053273313034220.
21. S. R. Wassell, S. Benito, Edge-length ratios between dual Platonic solids: a surprisingly new result involving the golden ratio, *Fibonacci Quarterly* 50 (2012) 144–154.
22. M. J. Wenninger, *Polyhedron Models*, Cambridge University Press (1971); *Dual Models*, Cambridge University Press (1983) (not read; see §5).